

MarginOptimizer

B2B Dynamic Alternative Flight & Yield Management System

Strands Agent Automated PNR Analysis

Key Metrics - **\$5.2B** | Global Dynamic Pricing Tech Market - **10K+** | Nightly PNR Scan Capacity - **+1-5%** | Marginal Net Profit Increase Per Booking - **100%** | Human-In-The-Loop Safety Record

TABLE OF CONTENTS

What This Document Covers

01 | The Cost Factor Extracting invisible profit from waiting tickets. **02 | The Solution Lifecycle** How the autonomous agent operates at 03:00 AM. **03 | Infrastructure & Scaling** Processing thousands of items without triggering API bans (SQS). **04 | The Approval Loop (HITL)** Slack Webhooks and operator-side security. **05 | Development Roadmap** From local validation to cloud dashboard deployment.

THE COST FACTOR

Creating Money From Inefficiency

The Scenario: A package tour is sold today for departure in 3 months. The flight cost locked today is \$1000. **The Opportunity:** In month 2, a competitor airline drops identical flights (same luggage, similar time) to \$800. Agencies miss this because humans cannot manually check 50,000 PNRs every day. **The Solution:** MarginOptimizer entirely automates this tracking using AI.

The ROI: Discovering just a \$20 saving on 10% of 50,000 yearly bookings = **\$100,000 pure Net Profit.**

SYSTEM ARCHITECTURE

AWS Serverless High-Throughput Engine

Service	Function in the Architecture
AWS EventBridge	Cron trigger launching the scan at 03:00 AM nightly
Amazon SQS	Queues 10,000 Bookings, protecting Traffics API from DDoS
Lambda (Worker)	Spawns Strands Agent to process singular queue items
API Gateway	Listens for human "Approve" clicks from Slack
DynamoDB/RDS	Logs every detected margin to generate BI dashboards

STRANDSTOOLING

The Margin Finders

- `use_traffics`
- **Target:** `/offers/{code}/alternativeFlights`
- **Execution:** Pings the Traffics spine to query identical alternative planes.
- `calculator_tool`
- **Execution:** Deterministically calculates (New Price - Old Price).
- **Integrity:** Prevents LLM mathematical hallucinations.
- `slack_tool`
- **Target:** Company operations channel.
- **Execution:** Dispatches interactive UI blocks (Approve/Reject) showing the found margin.

THE APPROVAL LOOP

Zero Unwanted Actions

The agent works entirely autonomously until a flight needs mutating. 1. The AI detects an \$80 margin on Booking ID 4410. 2. It sends a Slack notification: "Revise flight to SunExpress to save \$80?" 3. The human operator clicks **[APPROVE]**. 4. An AWS Webhook bypasses AI and directly triggers Traffics `/bookings/modify`. 5. Absolute data safety achieved.

INVESTMENT & ROADMAP

Execution Strategy

Phase 1: Validation (0-1 Mo) Prove the specific prompt logic isolates profit perfectly locally. **Phase 2: Cloud Farm (1-2 Mo)** Deploy AWS SQS queues. Tweak rate-limit handling for 429s. **Phase 3: Operational Link (2-3 Mo)** Integrate the Slack Webhooks and the direct Mutation scripts. **Phase 4: Big Data Returns (4+ Mo)** Roll out Amazon QuickSight automated C-Level "Yield Dashboards".

MarginOptimizer

Autonomous, Invisible, Highly Profitable

Confidential & Proprietary B2B Yield Optimization AI.